

$R = (A, B, C, D, E, F)$

$AB \rightarrow CDE$

$D \rightarrow F$

~~$AB \rightarrow DE$~~

$AB^+ = \{A, B, C, D, E, F\}$

$\{AB\}^+ = \{A, B, C, D, E, F\} = R$

→ only key (mark this in exam)

→ in 2NF

→ not in 3NF bc of $D \rightarrow F$ ← not prime attribute
↑
not superkey

→ not in BCNF bc not in 3NF

$F \in \{AB\}^+$

$F \notin \{A\}^+$

$F \notin \{B\}^+$

$S = (V, W, X, Y, Z)$

$V \rightarrow W, X, Y, Z$

~~$WZ \rightarrow V, X, Y$~~

$Y \rightarrow Z$

~~$FD \setminus \{WZ \rightarrow X, Y, Z\}$~~

check if rule is necessary

1 element

$\{V\}^+ = \{V, W, X, Y, Z\} = S \rightarrow \text{key!}$

$\{Y\}^+ = \{Y, Z\} \neq S$

2 elements

$\{W, Z\}^+ = \{W, Z, X, V, Y\} = S \rightarrow \text{key!}$

$\{W, Y\}^+ = \{W, Y, Z, X, V\} = S \rightarrow \text{key!}$

$\{Z, Y\}^+ = \{Z, Y\} \neq S$

3 elements

$\{X, Z, Y\}^+ = \{X, Z, Y\} \neq S \rightarrow \text{no more combinations}$

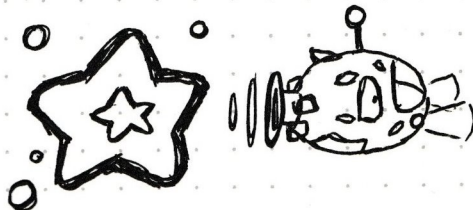
unnecessary bc can still be reached via $WZ \rightarrow V$

prime attributes = W, V, Y, Z non-prime = X

→ in 2NF

→ in 3NF

→ not in BCNF



DB contd 12/11/25

$Y = (\text{APHRODITE})$

$R \rightarrow O$

$O \rightarrow A$

$O \rightarrow H$

$O \rightarrow P$

$OR \rightarrow D$ $\{O\}^+ = OAHPRD$ shorten left side

$OR \rightarrow R$ $\{O\}^+ = OAHPRD$

$HP \rightarrow P$ trivial

$HPR \rightarrow D$ $\{R\}^+ = ROAHPRD$

Is D part of R without the FD that is being looked at?

$D \in \{R\}^+_{FD} = \{ROAHPRD\}$

$\hookrightarrow FD \setminus \{R \rightarrow D\}$

$\{ITE\}^+ \neq Y$

$\{ITER\}^+ = Y$ key

$\{ITEO\}^+ = Y$ key

④ 1) $AB^+ = ABCD$

$\rightarrow D$ depends just on B (key fraction)

2) $ABC^+ = ABCD = X'$ key

$BC^+ = ABCD = X'$ key

$\rightarrow$ BCNF, every LHS is superkey

3) $CDE^+ = CDEAB = V'$ key

$AE^+ = AEBCD \neq V'$

$CD^+ = CDEAB = V' \rightarrow$ key

$CD \rightarrow E$ violates 3NF

4) $A^+ = ABCD$

$AE^+ = AEBCD = Y' \rightarrow$ key prime attributes: AE

$\rightarrow$ every FD violates 3NF bc no right side is prime and $C \rightarrow D$ is not superkey

5) $ABC^+ = ABCD$

$AB^+ = ABCD$

$C^+ = CA \nrightarrow$ not BCNF conform

Does the attribute still point to all previous attributes without the FD that is being looked at? If not, then it can be excluded

e.g.: $HPR \rightarrow D$

$D \in \{R\}^+_{FD \setminus \{HPR \rightarrow D\}}$

short: $D \in \{R\}^+_{FD}$

File S6

$$R_1 = (AB)$$

$$R_2 = (CB)$$

DB contd. 11/12/25

$$R: \begin{array}{c} ABC \\ 123 \\ 423 \end{array}$$

$$R_3 \bowtie R_2 = \begin{array}{c} AC \ BC \\ 13 \ 23 \\ 45 \ 25 \end{array}$$

$$R_3 = \begin{array}{c} (AC) \\ 13 \\ 45 \end{array}$$

$$R_1 \bowtie R_2 \bowtie R_3 = \begin{array}{c} AB \ AC \ BC \\ 12 \ 13 \ 23 \\ 42 \ 45 \ 25 \end{array}$$

$R = \{AB C D E F G\}$

$AB \rightarrow C$

$B \rightarrow C$

$C \rightarrow D$

$DE \rightarrow F$

$AB \rightarrow F$

$AB C D E F G \rightarrow \emptyset$

augmented

$BD^+ = BCD$ $AB^+ = ABCDF$

$\{ABE\}^+ = ABCDEF \rightarrow$ superkey of R

$\rightarrow$ Problem: generates a superkey but not a key (sometimes)
 $\rightarrow$ unnecessary information

